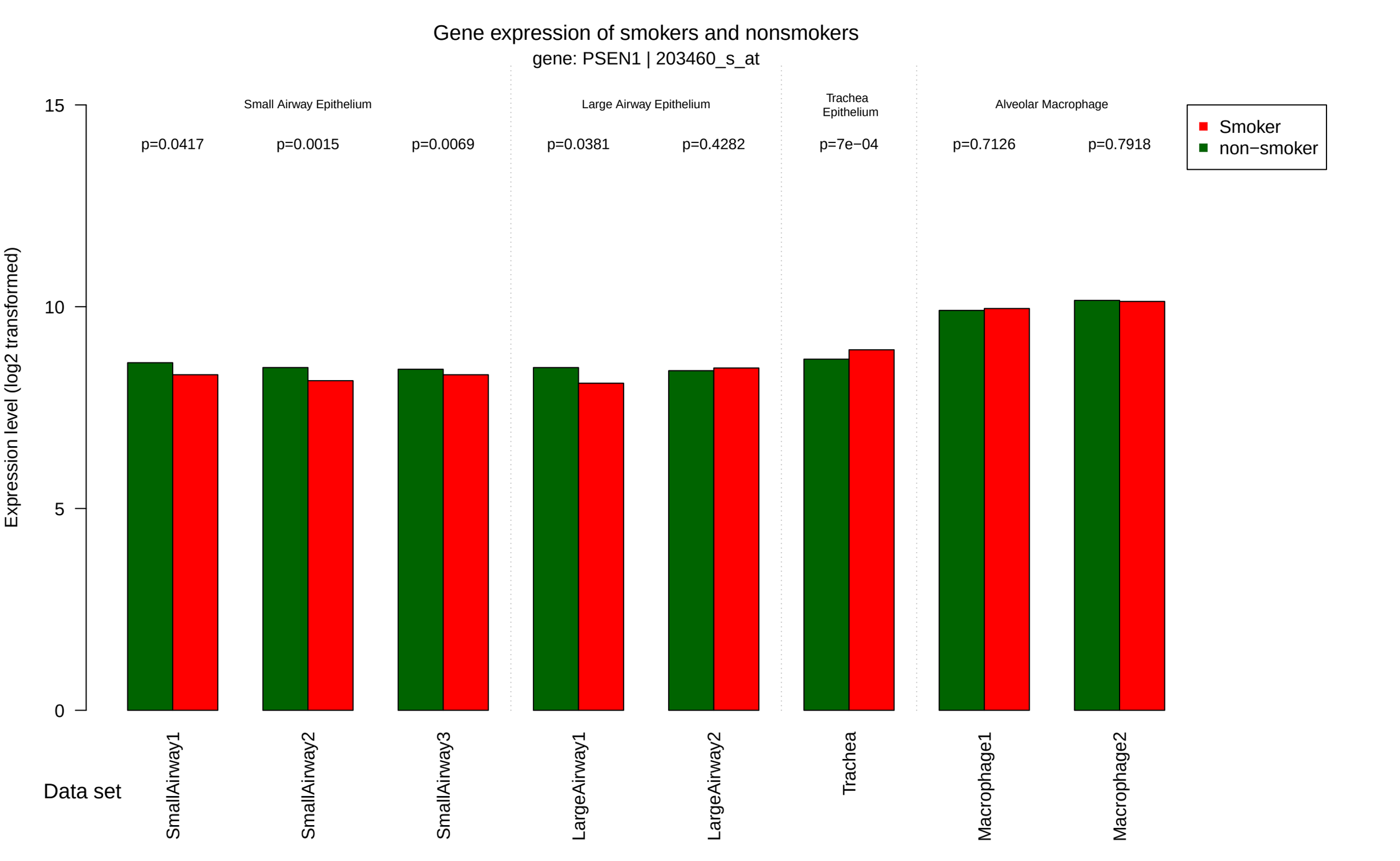
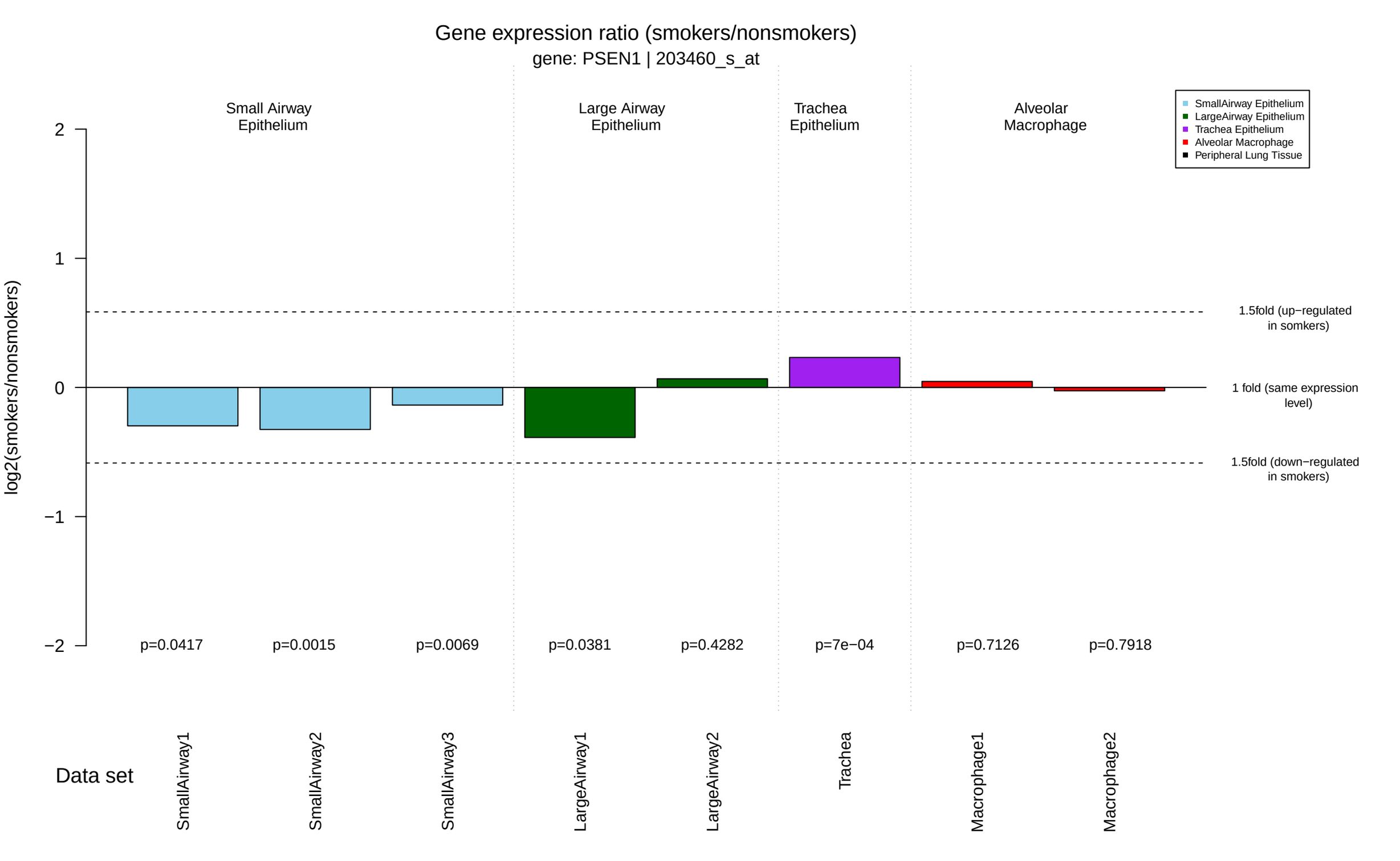
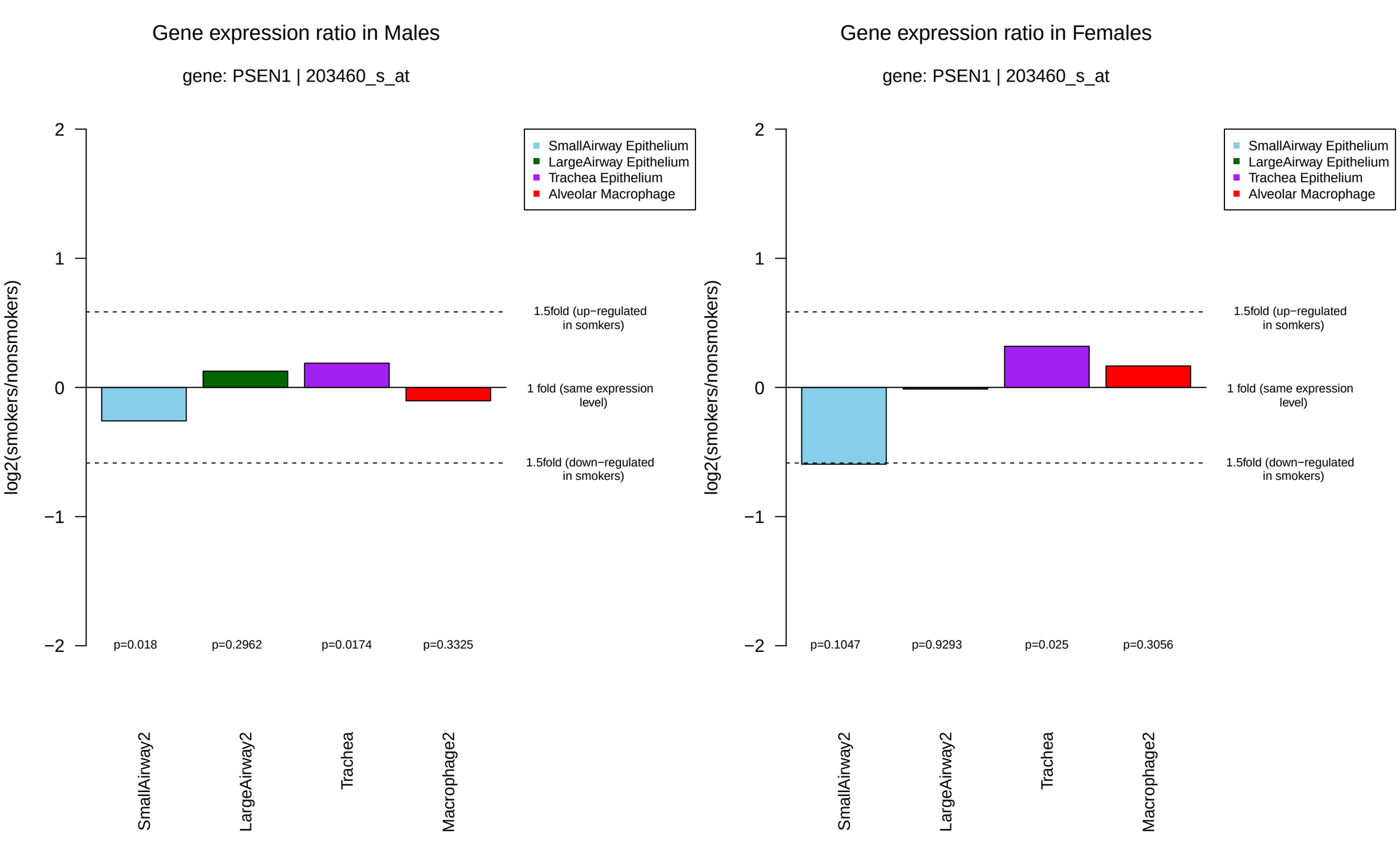
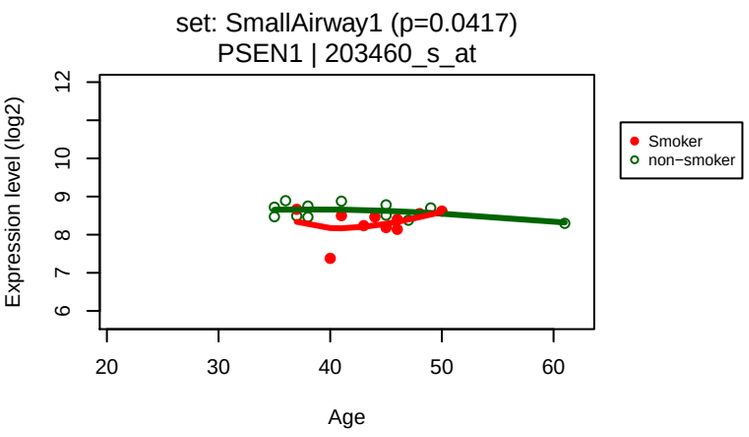
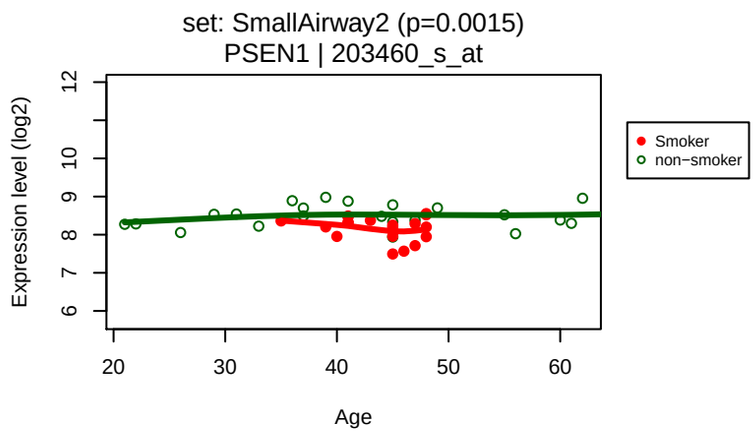
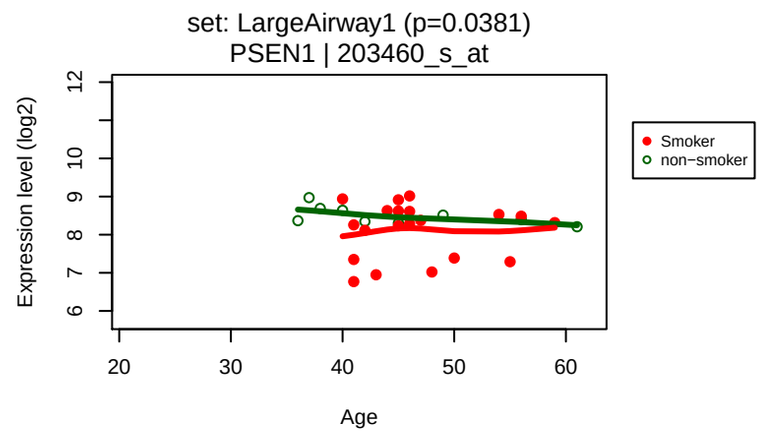

Gene expression of smokers and nonsmokers
gene: PSEN1 | 203460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0417
p=0.0015
p=0.0069
p=0.0381
p=0.4282
p=7e−04
p=0.7126
p=0.7918
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

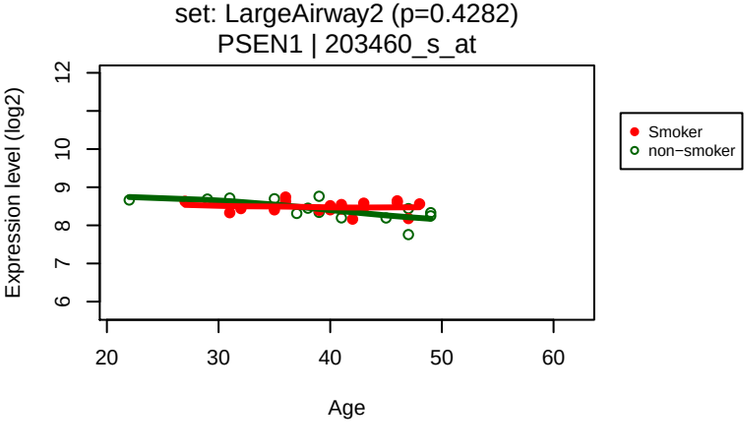

Gene expression ratio (smokers/nonsmokers)
gene: PSEN1 | 203460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.0015
p=0.0069
p=0.0381
p=0.4282
p=7e−04
p=0.7126
p=0.7918
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

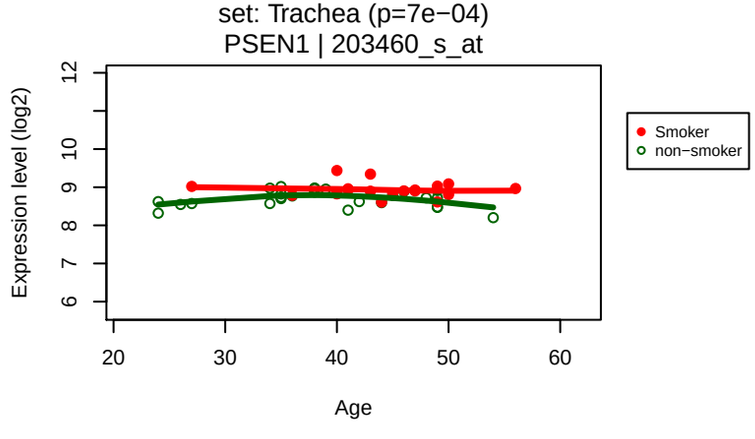

Gene expression ratio in Males
gene: PSEN1 | 203460_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.018
p=0.2962
p=0.0174
p=0.3325
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN1 | 203460_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1047
p=0.9293
p=0.025
p=0.3056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

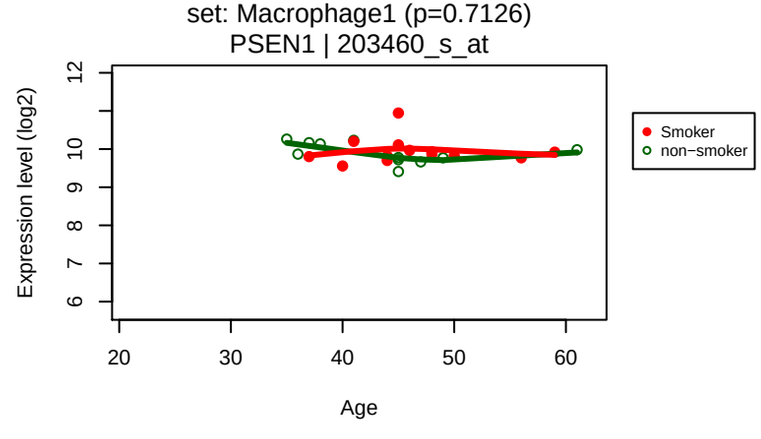

set: SmallAirway1 (p=0.0417)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker

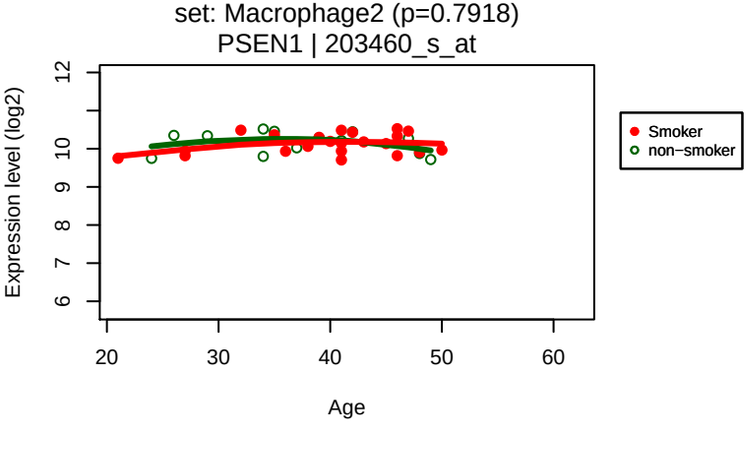

set: SmallAirway2 (p=0.0015)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker

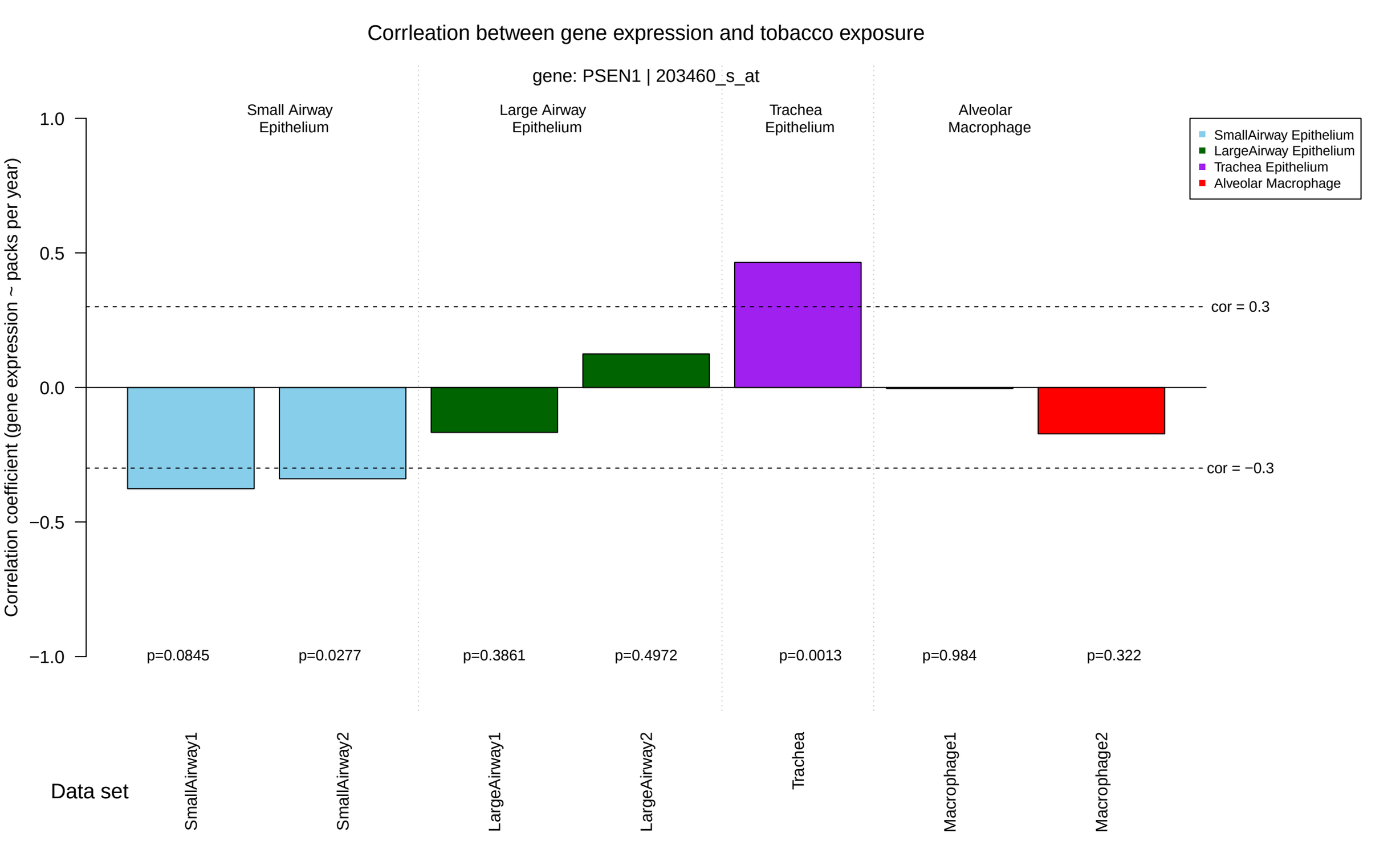

set: LargeAirway1 (p=0.0381)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4282)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7126)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7918)
PSEN1 | 203460_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN1 | 203460_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0845
p=0.0277
p=0.3861
p=0.4972
p=0.0013
p=0.984
p=0.322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2